

Measurement and estimation of plastic greenhouse reference evapotranspiration in a Mediterranean climate

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Abstract The standard FAO methodology for the determination of crop water requirements uses the product of reference evapotranspiration (ET_0) and crop coefficient values. This methodology can be also applied to soil-grown plastic greenhouse crops, which occupy extended areas in the Mediterranean basin, but there are few data assessing methodologies for estimating ET_0 in plastic greenhouses. Free-drainage lysimeters were used between 1993 and 2004 to measure ET_0 inside a plastic greenhouse with a perennial grass in Almería, south-eastern Spain. Mean daily measured greenhouse ET_0 ranged from values slightly less than 1 mm day^{-1} during winter to values of approximately 4 mm day^{-1} during summer in July. When the greenhouse surface was whitened from March to September (a common practice to control temperature), measured ET_0 was reduced by an average of 21.4%. Different methodologies to calculate ET_0 were checked against the measurements in the greenhouse without and with whitening. The methods that performed best in terms of

accuracy and statistics were: FAO56 Penman–Monteith with a fixed aerodynamic resistance of 150 s m^{-1} , FAO24 Pan Evaporation with a constant K_p of 0.79, a locally-calibrated radiation method and Hargreaves. Given the data requirements of the different methods, the Hargreaves and the radiation methods are recommended for the calculation of greenhouse ET_0 because of their simplicity.

Introduction

The greenhouse industry has expanded in many parts of the world (Enoch and Enoch 1999) and particularly throughout mild winter coastal areas in the Mediterranean Basin (Briassoulis et al. 1997; Pardossi et al. 2004). The southeast Spanish Mediterranean coastal area is one of the largest concentrations of greenhouses in the world, mainly dedicated to intensive vegetable production (Castilla and Hernández 2005). Most of these are low-cost structures covered with plastic film, without active climatic control systems and with drip-irrigated, soil-grown crops (Pérez-Parra et al. 2004). This agrosystem is now expanding rapidly in South and Central American countries (Pardossi et al. 2004). Greenhouse crops in Mediterranean areas are normally grown under sub-optimal climatic conditions, with temperatures above the optimum during the spring, summer and part of the autumn. For cooling the greenhouse atmosphere, natural ventilation is the normal practice (Baille et al. 2001), but it is generally not sufficient for alleviating the energy excess, and other cooling methods in combination with ventilation have to be used. Due to its low cost and high efficiency, whitewash or whitening the external plastic cover is the cooling method mostly used by farmers (in about 97% of the greenhouse surface in the region) (Pérez-Parra and Céspedes 2001). Whitening is normally maintained from

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March to September, except in greenhouses where zucchini, melon and watermelon crops are grown (Pérez-Parra and Céspedes 2001). Despite the widespread use of whitening for greenhouse cooling, investigations related to the influence of this practice on greenhouse microclimate and crop behaviour are scarce (Baille et al. 2001).

Simple transpiration models based on experimentally determined values of the ratio between crop transpiration and solar radiation have been developed for irrigation of greenhouse crops (de Villele 1974; de Graaf and van den Ende 1981). However, their use and accuracy in other greenhouse areas could be limited, as the proportion of greenhouse solar radiation used for transpiration in a mature crop without water limitation varies depending on the species, crop cycle and climate, greenhouse climate control systems, etc. (de Villele 1974; de Graaf and van den Ende 1981). In a Mediterranean greenhouse in the Southeast Spanish Mediterranean coastal area, the ratio between daily values of reference evapotranspiration (ET_o) and solar radiation (R_s) changed throughout the year, depending on the air temperature (Fernández 1993). Mechanistic transpiration models, based on the Penman–Monteith method (Monteith 1973), have also been developed for estimating greenhouse crop water requirements (Stanghellini 1987; Jolliet and Bailey 1992; Baille et al. 1994b; Boulard and Wang 2000; Montero et al. 2001; Möller et al. 2004; Medrano et al. 2005). However, their use is still quite limited as there is very limited information on the aerodynamic and canopy resistances of cropped surfaces that are required by these models.

The FAO K_c - ET_o methodology has been used worldwide to determine crop water requirements under open field conditions, but this methodology can be also applied to greenhouse crops (Bonachela et al. 2006). On the southeast Spanish Mediterranean coast, K_c values have been determined for the main greenhouse crops (Orgaz et al. 2005) and simple tables of daily crop water requirements, based on the single K_c - ET_o methodology, are available for farmers and technical advisers of greenhouse crops at <http://www.fundacioncajamar.com/estacion/agrdatos/riego/NecsRiego.htm>. In other Mediterranean greenhouse areas, the FAO24 Pan evaporation method has been recommended for estimating crop water requirements (Eliades 1988; Chartzoulakis and Drosos 1995). However, research on the measurement and modelling of greenhouse ET_o is scarce. De Villele (1974), who found a narrow relationship between observed greenhouse ET_o and solar radiation measurements, and Quaglietta and Zerbi (1986), using drainage and weighing lysimeters, respectively, measured ET_o in plastic greenhouses with perennial grass crops. Outdoors, ET_o has been measured under many different climatic conditions and a variety of methods have been proposed for its estimation (Allen et al. 1998).

Doorenbos and Pruitt (1977) proposed, in their FAO Irrigation and Drainage Paper No. 24 (FAO24), four different methods to calculate ET_o depending on the availability of weather data. Later on, Allen et al. (1998) in the FAO Irrigation and Drainage Paper No. 56 (FAO56), adopted a single ET_o calculation equation, the FAO56 Penman–Monteith method, which has been endorsed by the international scientific community because of the good results obtained in different regions of the world (e.g. Allen et al. 1989; Jensen et al. 1990; Smith et al. 1991; Choisnel et al. 1992; Ventura et al. 1999; Berengena and Gavilán 2005). Notwithstanding, there have been some modifications proposed for this method (Allen et al. 2006), and it appears to underestimate ET_o in semiarid and windy areas with high atmospheric evaporative demand (Steduto et al. 1996; Pereira et al. 1999; Ventura et al. 1999). In order to use the FAO K_c - ET_o methodology for greenhouse crops, measurements of greenhouse ET_o and assessment of the main greenhouse ET_o methods are required. Möller and Assouline (2007) observed that the daily ET_o under a cropped screenhouse in Israel, calculated from measured meteorological data (25 July to 15 November) following FAO56 methodology (Allen et al. 1998), ranged between 1 and almost 4 mm day⁻¹, and averaged 62% of the estimated outdoor value. The main reason for the reduction in ET_o , compared to outdoors, was the significant reduction in solar radiation inside the greenhouse (56% of open field conditions, on a seasonal basis), while the lower wind speeds inside contributed to a smaller extent. These authors also observed that the actual water use during the development stage of a sweet pepper crop, calculated from a daily water balance using lysimeter data, was similar to the computed daily crop water requirements, calculated with the FAO56 methodology.

The main objectives of this work were: (1) to measure the reference evapotranspiration inside a typical plastic greenhouse under Mediterranean conditions, and (2) to use these measurements to evaluate the performance of the main ET_o methods, with a particular emphasis on the FAO56 Penman–Monteith one.

Materials and methods

Site, experiments and management

Measurements of greenhouse reference evapotranspiration (ET_o) were carried out from 1993 to 2004 at the “Cajamar Foundation” research station (36° 48' N; 2° 3' W and 155 m a.s.l.) on the Almería coast, southeast Spain. The climate is Mediterranean with a mild winter (mean monthly temperature of 12.9°C at winter) and a mean annual precipitation of 220 mm.

ET_o measurements were conducted inside a typical Mediterranean greenhouse that was 24 m long and 20.5 m wide. A low-cost structure was used that was covered with plastic film (0.2 mm-thick thermal polyethylene sheet) with a symmetrical roof of 12.5% slope, E-W oriented, without heating equipment and passively ventilated by opening side panels and roof vents (Pérez-Parra et al. 2004). Vents were opened manually during most of the day during winter (except during the coolest period) and during the whole day for the rest of the year. The plastic film was changed in September 1993, August 1995, September 1997, October 2000 and July 2003. An artificially layered soil, typical of the region, was used consisting of a 30 cm layer of imported loamy soil placed above the naturally-occurring, gravelly sandy-loam soil. The upper limit of drained water content (field capacity) for the loamy soil layer was $0.31 \text{ cm}^3 \text{ cm}^{-3}$, and the lower limit (wilting point) was $0.11 \text{ cm}^3 \text{ cm}^{-3}$ (Bonachela et al. 2006).

The greenhouse had two drainage lysimeters (4 m length, 2 m width and 0.7 m depth) located on the southern side (Fig. 1) with the bottom and walls covered with a butyl rubber insulation sheet. The soil profile in the lysimeter reproduced that of the outside area described above to a depth of 0.6 m, with a layer of gravel placed on top of the butyl rubber sheet. The lysimeter soil depth was adequate as most of the root growth from greenhouse crops occurs within the loamy soil layer (Orgaz et al. 2005).

The greenhouse was uniformly planted with a perennial grass crop (*Cynodon dactylon* L.), which was maintained at

0.10–0.15 m height by regular mowing. This hot-season perennial grass showed good implantation and persistence under greenhouse conditions. Irrigation water of 0.4 dS m^{-1} electrical conductivity was applied weekly.

The plastic external cover was whitened with calcium carbonate at the usual concentration applied by growers in the area (about 40 g m^{-2}) to achieve greenhouse transmissivity values of between 40% and 50% of incoming global solar radiation from March to September in 2000, 2001, 2002, 2003 and 2004 years (Pérez-Parra and Céspedes 2001). When rainfall during these periods washed away part of the calcium carbonate, it was re-applied at a lower concentration to maintain the pre-established greenhouse transmissivity values.

Measurements

Greenhouse climate

An automatic agro-meteorological station (AWOS 7770, Thies Clima, Göttingen, Germany), mounted at 1.5 m height, and a class A evaporation pan were placed in the southern part of the greenhouse (Fig. 1). Air temperature and relative humidity were measured with a ventilated aspirated psychrometer (mod. 1.1130, Thies Clima, Göttingen, Germany), solar radiation with a pyranometer (mod. CM21, Kipp&Zonen, Delft, Netherlands) and wind velocity at 0.5 m with a cup anemometer (mod. Bestell, Thies Clima, Göttingen, Germany) with a resolution of 0.1 m s^{-1} and a precision of 0.3 m s^{-1} . Another automatic agro-meteorological station of similar characteristics and a class A evaporation pan were placed near the greenhouse, on bare land under open field conditions. Climatic data were measured every 10 s and these values were averaged every five 5 min and stored using a data logger (mod. DL-15, Thies Clima, Göttingen, Germany). Pan evaporation was measured daily at 8:00 a.m. The pan water was maintained clean and at a level that was 5–7 cm from the upper rim of the pan (Doorenbos and Pruitt 1977). Greenhouse radiation transmissivity was calculated as the ratio of measured daily values of global solar radiation inside and outside the greenhouse. Additional measurements of greenhouse air speed were taken from 18/12/2009 to 18/01/2010 to contrast those taken with the cup anemometer which, due to very low air speed inside the greenhouse, can be unreliable. They were carried out at a height of 2 m inside the greenhouse with a two dimensional sonic anemometer (mod. WindSonic, Gill Instruments Ltd, Hampshire, UK) and, simultaneously, in the outside agro-meteorological station with a cup anemometer (mod. Bestell, Thies Clima, Göttingen, Germany). Hourly air speed values (U_2) were calculated and a narrow regression relation between inside (U_{2i}) and outside (U_{2o})

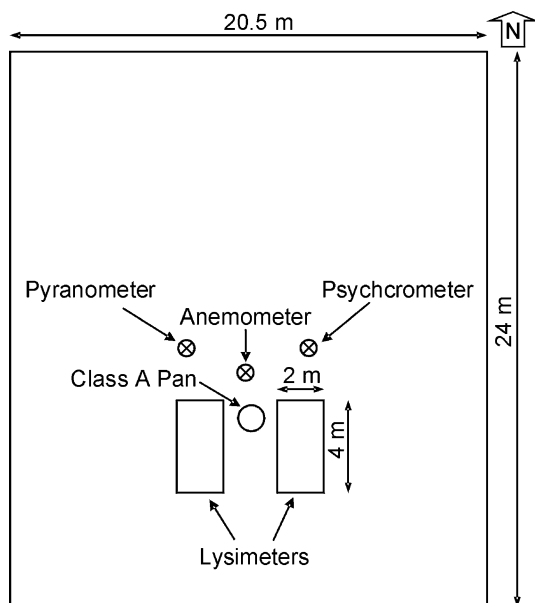


Fig. 1 Scheme of the experimental greenhouse with a perennial grass crop used for ET_o measurements from 1993 to 2004. Location of drainage lysimeters, class A pan and agro-climatic sensors. Almería, Spain

values was found (Wang et al. 1999) for the greenhouse with open vents (Eq. 1) and closed vents (Eq. 2). Using these equations, daily values of greenhouse air speed were recomputed for the 2 year series (1993–1999 and 2000–2004) and used in ET_o models. The range of greenhouse wind speed values measured with the sonic and the cup anemometer was similar.

$$U_{2i} = 0.0085U_{2o}^2 - 0.0055U_{2o} + 0.0403 \quad (1)$$

$$U_{2i} = 0.0038U_{2o}^2 - 0.009U_{2o} + 0.0055 \quad (2)$$

Reference evapotranspiration

Reference evapotranspiration (ET_o) was measured weekly in the lysimeters with the soil water balance approach, using the following equation:

$$ET_o = (SWC_{t0} - SWC_{t1}) + I - D \quad (3)$$

where ($SWC_{t0} - SWC_{t1}$) is the change in volumetric soil water content between two measurement dates; I and D are, respectively, the total volumes of applied irrigation water and collected drainage for the period under consideration. The volumetric soil water content in the lysimeters was measured weekly with a TDR system (TRASE 6005X1, Soil Moisture Corp., Santa Barbara, CA, USA). TDR measurements were made in four locations inside each lysimeter, at 0–45 cm deep. Drainage from lysimeters was collected daily and applied irrigation water was measured with a water meter. Mean daily ET_o values were calculated from weekly ET_o measurements.

The depth of irrigation water to be applied was determined using pan evaporation data. To ensure that ET_o was not limited by excessive or reduced soil water content, irrigation was adjusted to allow for some drainage from the lysimeter.

Reference evapotranspiration (ET_o) under open field conditions was calculated with the FAO56 methodology (Allen et al. 1998) using meteorological data from 2000 to 2004 measured in a nearby weather station from the Agroclimatic Information Network of Andalusia (Gavilán et al. 2006) at La Mojonera (37° 47' N; 2° 42' W, 142 m a.s.l.).

ET_o estimation methods

Greenhouse ET_o values were estimated with the FAO Penman–Monteith method (Allen et al. 1998), the FAO Penman, FAO Radiation and FAO Pan methods (Doorenbos and Pruitt 1977), and the Hargreaves method (Hargreaves and Samani 1985). Daily climatic data measured inside the greenhouse were used to calculate the daily ET_o with the computer program “REF-ET for windows, Version 2.0” (Allen 2000). Daily ET_o values were averaged over the corresponding week to obtain the mean daily ET_o .

A detailed description of methods, and of calculation and intermediate parameters can be found in Allen and Pruitt (1991), ASCE Manual no. 70 (Jensen et al. 1990), and FAO Manuals no. 24 (Doorenbos and Pruitt 1977) and no. 56 (Allen et al. 1998). The FAO56 Penman–Monteith equation (Allen et al. 1998) is:

$$ET_o = \frac{0.408\Delta(R_n - G) + \gamma \frac{900}{T+273} U_2 (e_s - e_a)}{\Delta + \gamma(1 + 0.34U_2)} \quad (4)$$

where ET_o is the reference evapotranspiration (mm day^{-1}); Δ is the saturation slope vapour pressure curve ($\text{kPa } ^\circ\text{C}^{-1}$); R_n is the net radiation at the grass surface ($\text{MJ m}^{-2} \text{day}^{-1}$); G is the soil heat flux ($\text{MJ m}^{-2} \text{day}^{-1}$); γ is the psychrometric constant ($\text{kPa } ^\circ\text{C}^{-1}$); U_2 is the mean daily wind speed at 2 m height (m s^{-1}); T is the mean daily air temperature ($^\circ\text{C}$); e_s is the air saturation vapour pressure (kPa); e_a is the actual vapour pressure (kPa). R_n and G were estimated following the procedure described in the FAO Manual no 56 (Allen et al. 1998). The FAO24 Penman equation (Doorenbos and Pruitt 1977) is:

$$ET_o = c \left[\left(\frac{\Delta}{\Delta + \gamma} \right) 0.408R_n + \left(\frac{\gamma}{\Delta + \gamma} \right) 2.7(1 + 0.01U_2)(e_s - e_a) \right] \quad (5)$$

where c is the adjustment factor calculated according to Allen and Pruitt (1991) from data of wind speed, relative air humidity and solar radiation. R_n was estimated following the procedure described by Doorenbos and Pruitt (1977). The FAO24 Radiation equation (Doorenbos and Pruitt 1977) is:

$$ET_o = b \times \left(\frac{\Delta}{\Delta + \gamma} \right) R_s - 0.3 \quad (6)$$

where b is the adjustment factor calculated according to Allen and Pruitt (1991) from daily data of wind speed and relative air humidity; R_s is the daily solar radiation at the grass surface (mm day^{-1}). The FAO24 Pan Evaporation equation (Doorenbos and Pruitt 1977) is:

$$ET_o = K_p \times E_o \quad (7)$$

where K_p is the pan coefficient that was calculated from data of mean daily relative humidity and wind speed, and fetch (Allen and Pruitt 1991); E_o is the pan evaporation (mm day^{-1}). The Hargreaves equation (Hargreaves and Samani 1985) is:

$$ET_o = 0.0023R_a \tau (T_{\max} - T_{\min})^{1/2} (T + 17.8) \quad (8)$$

where R_a is the extraterrestrial radiation (mm day^{-1}); τ is the greenhouse transmissivity which is the ratio between the inside and the outside solar radiation; T , $T_{\max}$ and $T_{\min}$ are the mean, maximum and minimum greenhouse air temperatures

(°C), respectively. Finally, the equation of a locally-calibrated radiation method (Bonachela et al. 2006) is:

$$\text{Julian days (JD)} \leq 220 \quad \text{ET}_o = (0.288 + 0.0019\text{JD})R_o\tau \quad (9)$$

$$\text{Julian days (JD)} \leq 220 \quad \text{ET}_o = (1.339 - 0.00288\text{JD})R_o\tau \quad (10)$$

where R_o is the daily solar radiation outside the greenhouse (mm day^{-1}). Comparisons between estimated (E) and measured (O) daily greenhouse ET_o values were carried out by simple linear regression. The regression through the origin was selected to evaluate the goodness of the fit between ET_o estimates and lysimeter measurements, as both values should theoretically approach the origin when the actual ET_o is zero (Allen et al. 1989). Additionally, the following statistics were calculated as described by Willmott (1982): root mean square error (RMSE, mm day^{-1}), mean bias error (MBE, mm day^{-1}) and relative error (RE, %).

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^n (E_i - O_i)^2}{n}} \quad (11)$$

$$\text{MBE} = \frac{\sum_{i=1}^n (E_i - O_i)}{n} \quad (12)$$

$$\text{RE} = \frac{\text{MBE}}{\bar{O}} \times 100 \quad (13)$$

where n is the total observation number and $\bar{O}$ is the mean of measured values. Estimated and measured daily ET_o values, averaged weekly, were compared throughout each week of the year from 1993 to 1999 in the greenhouse without whitening and from 2000 to 2004 in the whitened greenhouse. When statistical parameter adjustment was used for improving the performance of an ET_o method, different data sets were used for parameter calibration and model validation. In the FAO24 Pan and the locally-calibrated radiation methods, measured ET_o data from 1993 and 1994 were used for calibration and data from 1995 to 2004 for validation.

Results

Greenhouse Climate

Figure 2 shows the evolution of daily values (weekly averages) of solar radiation, and relative humidity (from 1993 to 2004) inside a grass-cropped greenhouse with (from 2000 to 2004) and without whitening (from 1993 to 1999), and outdoors (from 1993 to 2004). Figure 2 also shows mean daily values of the temperature gradient between greenhouse and outside air. Daily outdoor values

(weekly averages) of solar radiation, air temperature and relative air humidity were similar throughout the year for both series studied: from 1993 to 1999 and from 2000 to 2004 (data not shown). The mean daily integral of solar radiation outside the greenhouse ranged between $8 \text{ MJ m}^{-2} \text{ day}^{-1}$ in winter and $26 \text{ MJ m}^{-2} \text{ day}^{-1}$ in summer (Fig. 2a). The mean value of greenhouse radiation transmission was 60.9% for the years without whitening, and 53.4% for the years with whitening (60.8% and 48.1%, respectively, when only the whitening period of the year was considered). Mean daily values of greenhouse air temperature were, on average, 1.0°C and 1.4°C higher than outdoors throughout the year and for the whitening period of the year, respectively (Fig. 2b). The differences were smaller in the winter period when the inside night-time air temperatures were similar to the outdoor values (Kittas 1995; López 2003) or even lower under clear sky and low wind conditions (Papadakis et al. 2000). Whitening reduced the air temperature gradient as the greenhouse air temperature was, averaged for the whitening period of the year, 0.5°C higher than outdoors (Fig. 2b). The mean daily air relative humidity was higher inside the greenhouse (values between 65% and 80% for most of the year) than outdoors (values between 60% and 70%), especially in autumn, winter and spring (Fig. 2c). On the whole, whitening produced slightly higher relative humidity values inside the greenhouse in summer (Fig. 2c), but the mean vapour pressure deficit during the whitening period was similar in the greenhouse with (1.2 kPa) and without whitening (1.2 kPa), and slightly higher than outdoors (1.0 kPa). The mean wind speed outdoors ranged between 1.5 and 3 m s^{-1} for most of the year, and it was drastically reduced inside the greenhouse, with values of 0.1 to 0.3 m s^{-1} for most of the year.

Greenhouse reference evapotranspiration (ET_o) and pan evaporation (E_o)

Figure 3 shows the seasonal evolution of mean daily values of ET_o and E_o , measured inside a greenhouse without (1993 to 1999) and with whitening (2000 to 2004).

The mean daily greenhouse ET_o ranged from values slightly lower than 1 mm day^{-1} in December–January up to values slightly higher than 4 mm day^{-1} at the beginning of July (Fig. 3a). ET_o values outdoors, computed after Allen et al. (1998) from measured meteorological data in a nearby weather station, ranged from values of approximately 1.5 mm day^{-1} in winter up to values of approximately 6.5 mm day^{-1} at summer (data not shown). Whitening reduced ET_o values by 21.4% averaged over the whitened period (Fig. 3a), and the maximum values were slightly higher than 3 mm day^{-1} . Mean daily greenhouse

Fig. 2 Year evolution of daily values, weekly averaged, of solar radiation (a), temperature gradient between greenhouse and outside air (b) and relative humidity (c), measured outside (from 1993 to 2004) and inside a greenhouse without (from 1993 to 1999) and with whitening (from 2000 to 2004)

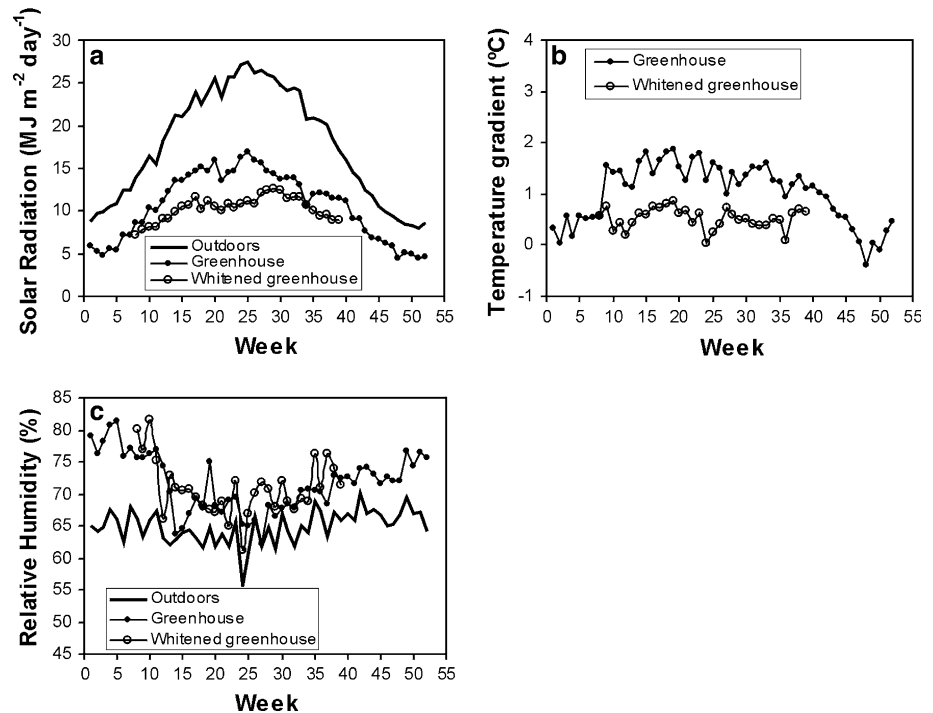
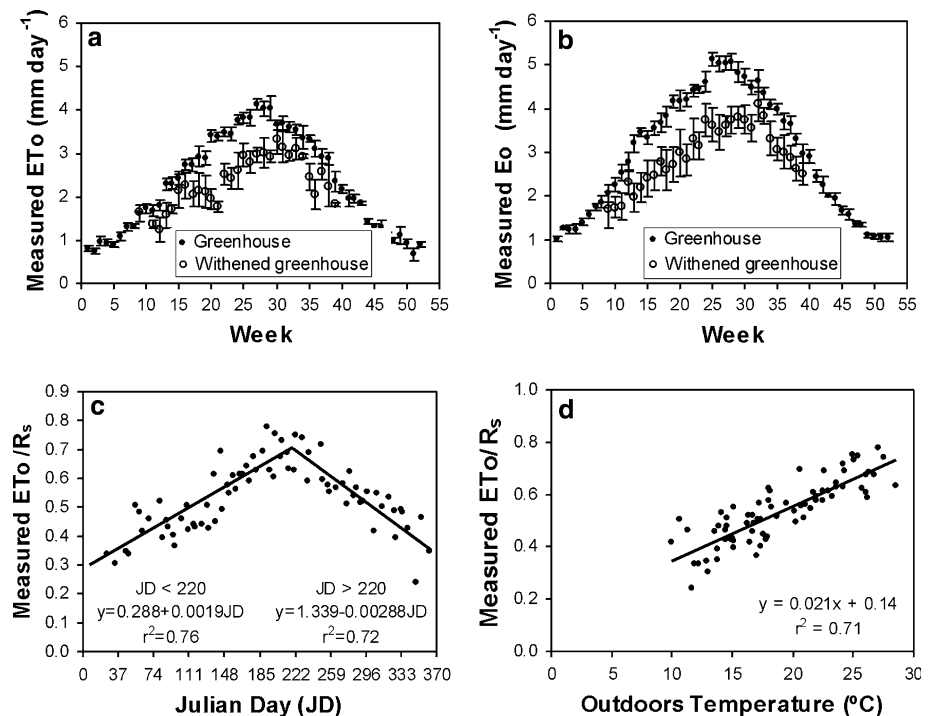


Fig. 3 Year evolution of daily values, weekly averaged, of ET_o (a) and E_o (b), measured inside a plastic greenhouse with (from 2000 to 2004) and without whitening (from 1993 to 1999). Bars indicate twice the standard error of measurements. c Year evolution of the ratio between of daily values, weekly averaged, of ET_o and solar radiation (R_s) inside a plastic greenhouse without whitening; d Relationship between this ratio and the outside air temperature. Each point from (c) and (d) corresponds with a week throughout the years 1993 and 1994



E_o values presented similar seasonal patterns to ET_o values (Fig. 3b). The mean daily greenhouse E_o varied between 1 mm day^{-1} in December–January and values slightly higher than 5.0 mm day^{-1} at the beginning of July, while whitening reduced E_o values by 23.4% over the whitened period. For the whole period of measurement, greenhouse ET_o values were 25% (the years with whitening) and

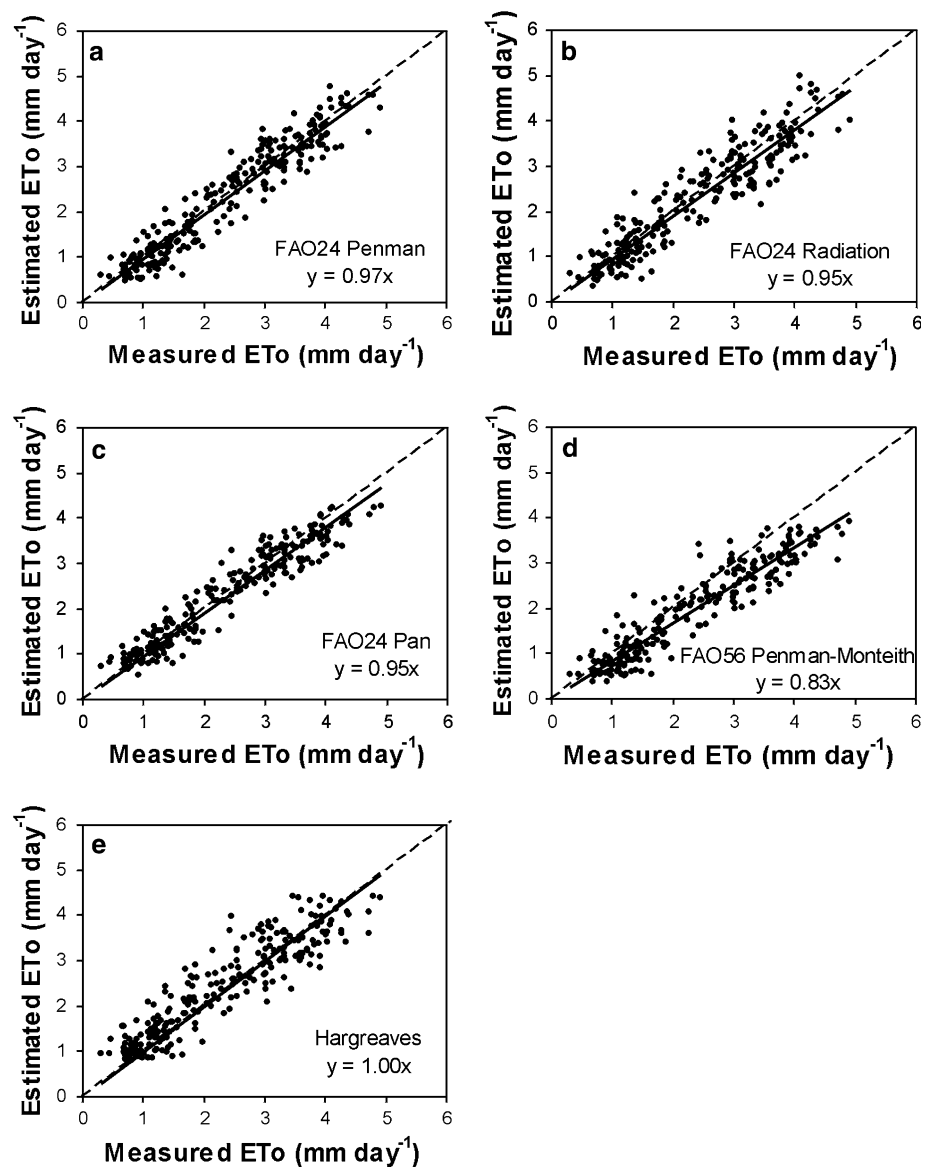
22.7% (the years without whitening) lower than E_o values. Outdoors, the mean daily E_o ranged between values slightly higher than 2.0 mm day^{-1} in December–January and values slightly higher than 9.0 mm day^{-1} in July (data not shown).

The higher standard error obtained for most ET_o and E_o values in the years when the greenhouse was whitened

(mean values of 0.25 and 0.34 mm day⁻¹, respectively) (Fig. 3a, b), compared to years without whitening (mean values of 0.13 and 0.14 mm day⁻¹, respectively), may be partly attributed to the lower number of observations during the years of whitening (3 to 4 depending on the week), compared to the years without whitening (5 to 7 depending on the week). However, the variation in the transmissivity in the whitened greenhouse must have also contributed to the larger standard error of ET_o and E_o values.

Figure 3 also shows the year evolution of the ratio between daily values, averaged weekly, of ET_o and solar radiation inside a greenhouse without whitening throughout 1993 and 1994. This ratio was related to air temperature (Fig. 3d) and changed throughout the year (Fig. 3c): values increased from around 0.35 in mid-winter to 0.75 in mid-summer and they decreased hereafter.

Fig. 4 Comparison between mean daily values of estimated versus measured ET_o values in a plastic greenhouse without whitening. ET_o values were estimated by FAO24 Penman (a), FAO24 Radiation (b), FAO24 Pan (c), FAO56 Penman–Monteith (d) and Hargreaves (e) methods. Each point corresponds with a daily ET_o value, averaged over a week, throughout a particular year from 1993 to 1999



Evaluation of ET_o methods

Figure 4 presents the comparison of mean daily values of estimated versus measured ET_o in a plastic greenhouse without whitening. The FAO24 Penman, FAO24 Radiation and FAO24 Pan methods accurately estimated measured ET_o values, as most estimated data are closely distributed around the 1:1 line (Fig. 4a, b and c, respectively). These three methods also gave good statistical indices (Table 1). Measured ET_o values were largely overestimated (66% on average) by the Hargreaves method when the original equation (Hargreaves and Samani 1985) was used (data not shown), but they accurately tracked measured ET_o when the extraterrestrial radiation was multiply by greenhouse radiation transmissivity (Fig. 4e, Table 1). By contrast, the FAO56 Penman–Monteith method clearly underestimated

Table 1 Summary of statistics from the comparison between estimated (E , mm day^{-1}) and measured (O , mm day^{-1}) values of mean daily greenhouse ET_o with (2000–2004) and without whitewash (1993–1999)

ET_o method	N	O	E	E/O	Slope	r^2	RMSE	MBE	RE
Greenhouse without whitening									
FAO24 Penman	235	2.36	2.30	0.97	0.97	0.98	0.36	−0.07	−3.0
FAO24 Radiation	236	2.36	2.27	0.96	0.95	0.97	0.45	−0.10	−4.3
FAO24 Pan	236	2.36	2.28	0.97	0.95	0.98	0.34	−0.08	−3.2
FAO56 Penman–Monteith	235	2.36	1.98	0.84	0.83	0.97	0.56	−0.39	−16.4
Hargreaves (greenhouse transmissivity)	237	2.36	2.46	1.04	1.00	0.97	0.47	0.09	3.7
Greenhouse with whitening									
FAO24 Penman	82	2.34	2.05	0.88	0.88	0.98	0.42	−0.26	−11.0
FAO24 Radiation	82	2.34	2.08	0.89	0.88	0.97	0.44	−0.23	−10.0
FAO24 Pan	83	2.34	2.34	1.00	1.00	0.98	0.34	0.03	1.4
FAO56 Penman–Monteith	82	2.34	2.10	0.90	0.88	0.98	0.40	−0.22	−9.2
Hargreaves (greenhouse transmissivity)	82	2.34	2.27	0.97	0.95	0.97	0.44	−0.06	−2.6

N total number of observations; slope of the regression equation, r^2 coefficient of determination, $RMSE$ root mean square error (mm day^{-1}), MBE mean bias error (mm day^{-1}), RE relative error (%)

measured ET_o values, the underestimation being 17% on average (Fig. 4d, Table 1).

ET_o methods were also evaluated when the greenhouse was whitened (Fig. 5). Principally, the FAO24 Pan, and also the Hargreaves method presented the best agreement between estimated and measured values (most estimated data are closely distributed around the 1:1 line and the slopes of the regression equations are not significantly different from unity). They also presented, in general, the best statistical indices (Table 1). By contrast, measured ET_o data were slightly underestimated by the FAO24 Radiation (12% on average), the FAO24 Penman (12%) and the FAO56 Penman–Monteith (12%) methods (Table 1).

Discussion

Greenhouse ET_o

The mean daily greenhouse ET_o , measured over weekly periods in a plastic greenhouse with a perennial grass crop, ranged from 0.7 to 4.1 mm day^{-1} (Fig. 3). This range of values is similar to that calculated by Möller and Assouline (2007) using the FAO56 Penman–Monteith equation (Allen et al. 1998) in a cropped screenhouse at Israel, but it is much lower than outdoors ET_o values measured in areas of similar climate (Berengena and Gavilán 2005; Lecina et al. 2003) or was calculated for open fields in a nearby weather station. The plastic greenhouse reduced solar radiation inside by 39% on average (Fig. 2), and substantially reduced wind speed inside with values being approximately or lower than 0.1–0.3 m s^{-1} for most of the

year. As an overall assessment, solar radiation appeared to be the meteorological variable determining most of the greenhouse ET_o . Similar assessments were made by De Villele (1974) and Möller and Assouline (2007). However, the ratio between ET_o and R_s values inside a Mediterranean greenhouse changed throughout the year (Fig. 3c) and these changes were related with the air temperature (Fig. 3d). This could be related to the relationship between solar radiation and net radiation, which varies with the greenhouse temperature. Therefore, it is not recommended to use this type of relationships as an accurate predictive tool for estimating transpiration rates in low-cost Mediterranean greenhouses. When the greenhouse was whitened with calcium carbonate from March to September, a common practice for several horticultural crops in the region, the mean daily solar radiation inside was reduced by 18.3% (Fig. 2a) and the mean daily greenhouse ET_o by 21.4% (Fig. 3a), both averaged over the whitened period and compared to the years without whitening. Möller and Assouline (2007) found that the daily ET_o under a screenhouse was 38% lower than the outdoor value, both computed from measured meteorological data after Allen et al. (1998), whereas the solar radiation inside was reduced by 44%. The relatively higher ET_o reduction observed in the whitened plastic greenhouse, compared to the screenhouse (Möller and Assouline 2007), could be mainly ascribed to the drastic wind speed reduction inside the greenhouse, which is much higher than in the screenhouse.

Whitening also produced a greenhouse cooling effect as the air temperature was only 0.5°C higher inside the greenhouse than outside, on average (Fig. 2b). Baille et al. (2001) in a glasshouse growing a rose crop in the coastal

area of eastern Greece, also found that glasshouse whitening was an efficient cooling method, producing positive effects on both microclimate (air temperature and humidity values were close to outside conditions) and crop behaviour (crop transpiration was higher after greenhouse whitening because the rose crop was less water stressed).

Greenhouse ET_o methods

The reference evapotranspiration methods recommended by the FAO24 (Doorenbos and Pruitt 1977) and the FAO56 (Allen et al. 1998) manuals, and the ASCE (Jensen et al. 1990) were evaluated comparing estimated versus measured greenhouse ET_o data. The temperature-based Hargreaves equation accurately estimated measured greenhouse ET_o without whitening (Fig. 4e; Table 1) and presented a slight

underestimation of 5% with whitening (Fig. 5e; Table 1) when the extraterrestrial radiation was multiplied by the measured greenhouse radiation transmissivity for considering the actual radiation entering the greenhouse. This method, an empirical approach widely used and evaluated under open field conditions (Allen et al. 1998; Itenfisu et al. 2003; Gavilán et al. 2006) that only requires temperature and greenhouse radiation transmissivity data, could be easily applied for determining greenhouse ET_o .

The FAO24 Pan method also accurately estimated the measured greenhouse ET_o with whitening (Fig. 5c; Table 1) and presented a slight underestimation of 5% without whitening (Fig. 4c; Table 1). Data of class A pans, but installed outside the greenhouse, had been previously used by Eliades (1988) and Chartzoulakis and Drosos (1995) to estimate greenhouse crop water requirements in

Fig. 5 Comparison between mean daily values of estimated versus measured ET_o values in a plastic greenhouse with whitening. ET_o values were estimated by FAO24 Penman (a), FAO24 Radiation (b), FAO24 Pan (c), FAO56 Penman–Monteith (d) and Hargreaves (e) methods. Each point corresponds with a daily ET_o value, averaged over a week, throughout a particular year from 2000 to 2004

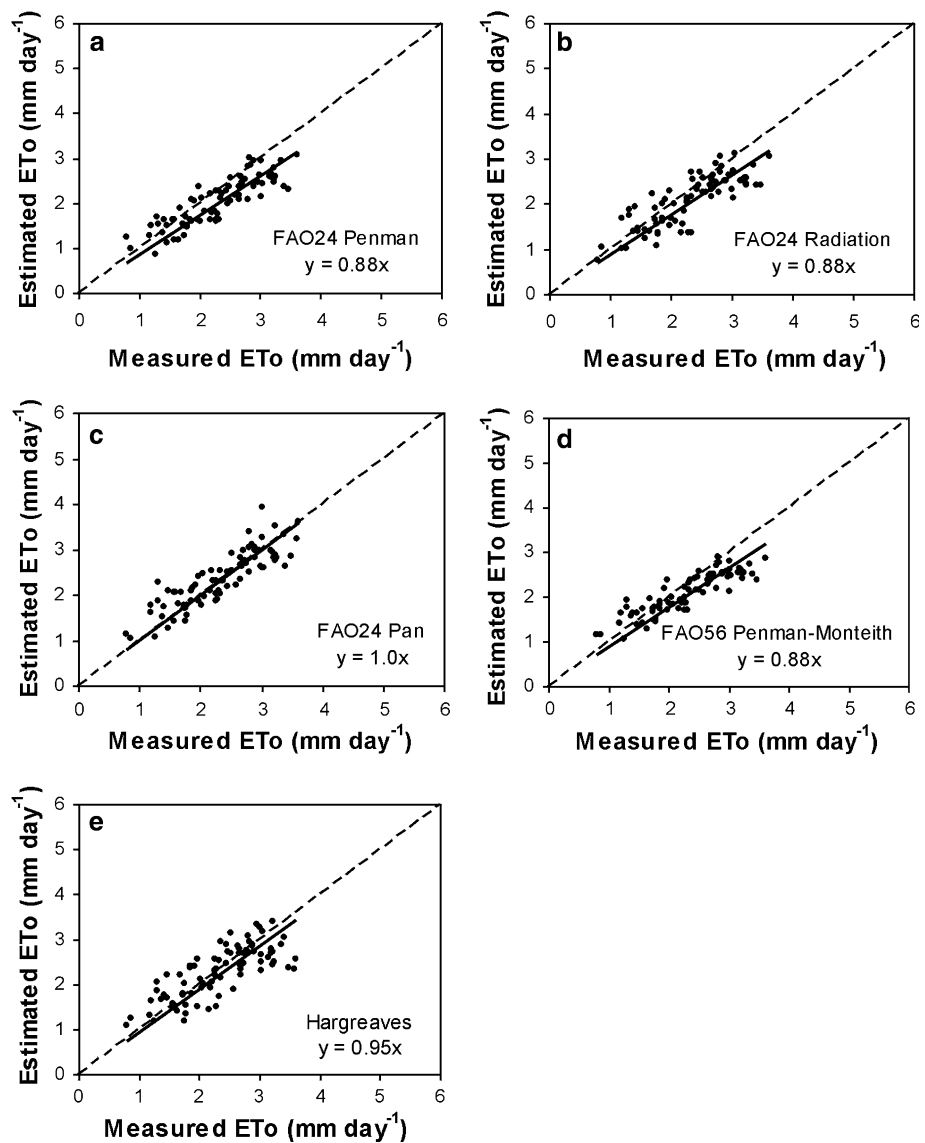
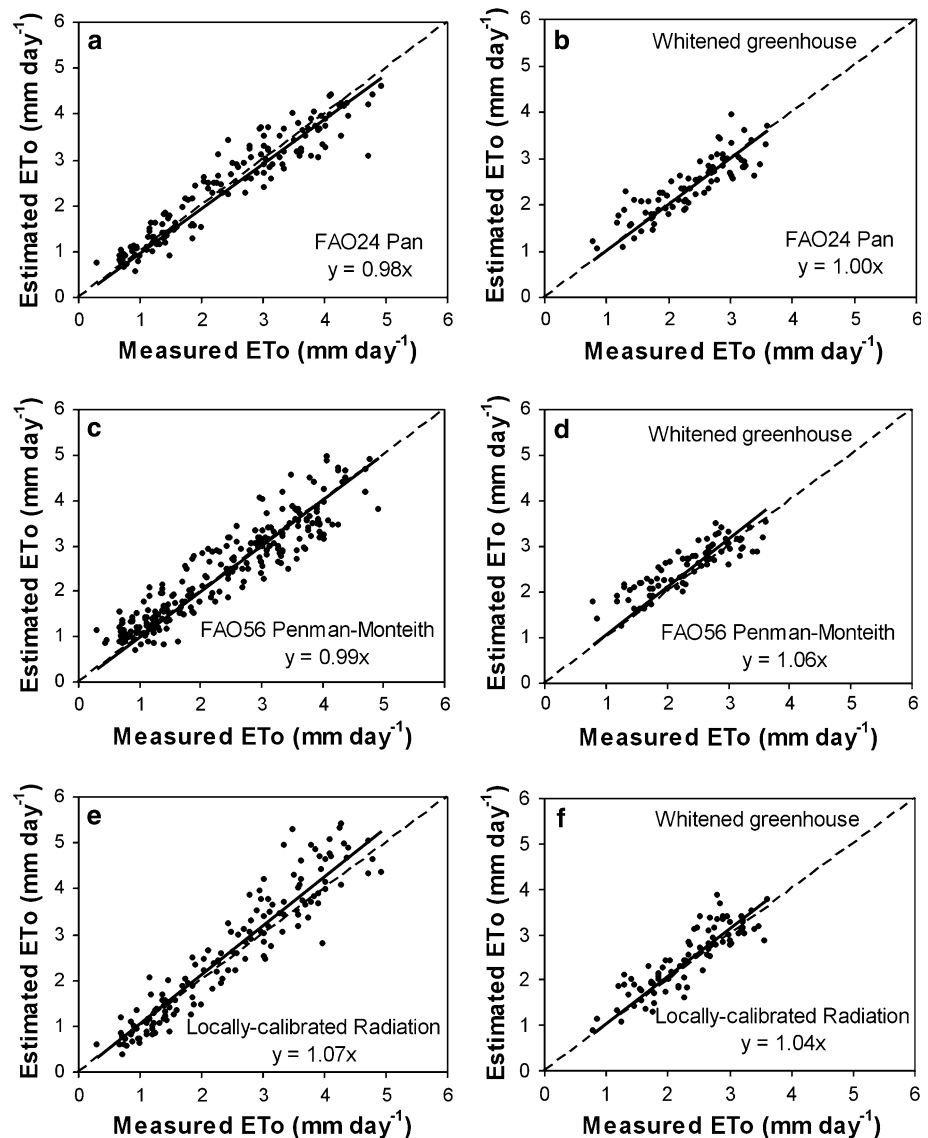


Fig. 6 Comparison between mean daily values of estimated and measured ET_o values in a plastic greenhouse with (from 2000 to 2004) and without whitening (from 1993 to 1999 for (c); and from 1995 to 1999 for a and e. ET_o values were estimated by FAO24 Pan method using a constant K_p value of 0.79 (a and b), FAO56 Penman–Monteith using a fixed aerodynamic resistance value of 150 s m^{-1} (c and d) and a radiation method locally calibrated (e and f). Each point corresponds with a daily ET_o value, averaged over a week, throughout a particular year



two Mediterranean areas. The K_p value, estimated from data of mean daily relative humidity and wind speed, and fetch (Allen and Pruitt 1991), maintained rather constant values throughout the year in the greenhouse with and without whitening with mean values 0.79 and 0.77, respectively. A K_p value of 0.79 was also experimentally determined as the slope of the linear regression forced through the origin (the intercept did not differ statistically from zero) between measured greenhouse data of ET_o and E_o throughout 1993 and 1994. Using this K_p value, the accuracy of the FAO24 Pan method improved for the greenhouse without whitening (2% underestimation, Fig. 6a; Table 2) and stayed very high with whitening (Fig. 6b; Table 2). This method, however, cannot be directly applied to greenhouses whose radiation transmission values differ from those in which the class A pan is installed (e.g. greenhouses whitened with different

concentrations of calcium carbonate). Besides, this method requires the maintenance of a perennial grass crop in a greenhouse.

The FAO56 Penman–Monteith method, recommended as the standard method for ET_o estimations (Allen et al. 1998), clearly underestimated the measured greenhouse ET_o , especially without whitening (Figs. 4d and 5d; Table 1). This method incorporates an equation for calculating the aerodynamic resistance (r_a) as a function of wind velocity. Because measured daily wind velocity values inside the greenhouse mostly oscillated between 0.01 and 0.2 m s^{-1} , the calculated r_a values for the grass-cropped greenhouse oscillated between 20,800 and $2,080 \text{ s m}^{-1}$. These values are much higher than those observed for several greenhouse crops, that oscillated between 100 and 500 s m^{-1} (Bailey et al. 1993; Baille et al. 1994a; Katsoulas et al. 2001). Based on Bailey et al. (1993) who estimated r_a

Table 2 Summary of statistics from the comparison between estimated (E , mm day^{-1}) and measured (O , mm day^{-1}) data of mean daily greenhouse ET_o with (2000–2004) and without whitewash (for

FAO56 Penman–Monteith method from 1993 to 1999, and for FAO24 Pan and Locally-calibrated Radiation methods from 1995 to 1999)

ET_o method	N	O	E	E/O	Slope	r^2	RMSE	MBE	RE
Greenhouse without whitening									
FAO56 Penman–Monteith ($r_a = 150 \text{ s m}^{-1}$)	237	2.37	2.42	0.98	0.99	0.97	0.43	0.05	2.1
FAO24 Pan ($K_p = 0.79$)	148	2.54	2.39	1.01	0.98	0.98	0.37	−0.01	−0.4
Locally-calibrated radiation	145	2.54	2.56	1.09	1.07	0.98	0.49	0.14	5.9
Greenhouse with whitening									
FAO56 Penman–Monteith ($r_a = 150 \text{ s m}^{-1}$)	83	2.34	2.49	1.06	1.06	0.98	0.39	0.20	8.5
FAO24 Pan ($K_p = 0.79$)	83	2.34	2.35	1.00	1.00	0.98	0.34	0.04	1.8
Locally-calibrated radiation	83	2.34	2.45	1.05	1.04	0.98	0.38	0.13	5.6

N total number of observations; slope of the regression equation, r^2 coefficient of determination, $RMSE$ root mean square error (mm day^{-1}), MBE mean bias error (mm day^{-1}), RE relative error (%)

values of a greenhouse crop as a function of leaf area index and leaf dimensions, we can assume for the greenhouse perennial grass crop, maintained at 0.1–0.15 m height by regular mowing, practically constant values of leaf area index and leaf dimensions, and, therefore, constant r_a values over the season. Assuming constant r_a values over the season of 100, 150, 200 and 250 s m^{-1} , respectively, the greenhouse ET_o was estimated with the FAO56 Penman–Monteith method. In general, the goodness of fit of ET_o estimates clearly improved with these r_a values, and the best performance was found for a r_a value of 150 s m^{-1} for both cases, of the greenhouse with and without whitening (Fig. 6c and d, Table 2). With the introduction of a fixed r_a value of 150 s m^{-1} , the FAO56 Penman–Monteith method predicted greenhouse ET_o correctly, although it requires detailed greenhouse microclimatic measurements, which are not always available, while it did not improve the prediction capability of more simple methods (Table 2).

The FAO24 Radiation and the FAO24 Penman methods performed in a similar way. Measured greenhouse ET_o values were accurately predicted without whitening, but they were underestimated with whitening (Figs. 4 and 5; Table 1). The correction parameter values of b for the FAO24 Radiation and c for the FAO24 Penman method were rather stable in the whitened greenhouse with mean values of 0.78 and 0.96, respectively. However, in the greenhouse without whitening b and c values were stable and similar to those observed for whitening conditions for the winter period, but they increased progressively in spring, reaching maximum values at summer (14% more for b and 11% more for c than those calculated under whitening conditions) and decreased progressively in autumn. These methods, therefore, would require a local calibration of the correction parameters for accurate ET_o estimations in whitened greenhouses. Taking into account these results and that radiation methods work well to estimate accurately ET_o in humid climates, where the

aerodynamic term is relatively small (Allen et al. 1998; Jensen et al. 1990), a locally-calibrated radiation method introduced by Bonachela et al. (2006) was also evaluated. Although this method slightly overestimated the measured ET_o data in the greenhouse with and without whitening (Fig. 6e and f; Table 2), it behaved, in general, in a similar way than the modified FAO56 Penman–Monteith method (Fig. 6c and d; Table 2). Given that most of the flexible plastic films used as greenhouse covering materials do not differ much in their radiation transmission values (Papadakis et al. 2000) and considering that greenhouse ET_o methods were assessed under a wide range of climatic (from 1993 to 2004) and greenhouse management conditions (with and without whitening, open and closed vents, etc.), we believe that the results of this research can be, at least partially, applied to other areas of plastic greenhouses without active climate control systems located in regions with a Mediterranean climate.

Conclusions

In a semi-arid Mediterranean area the mean daily greenhouse ET_o ranged from values slightly lower than 1 mm day^{-1} at winter up to values slightly higher than 4 mm day^{-1} at July, and when the greenhouse was whitened with a moderate calcium carbonate concentration from March to September it was reduced by 21.4%. Moreover, the relationship between greenhouse ET_o and solar radiation changed throughout the year.

Mean daily greenhouse ET_o was accurately estimated by the Hargreaves, FAO56 Penman–Monteith (using a fixed r_a value of 150 s m^{-1}), FAO24 Pan (using a fixed K_p value of 0.79) and a locally-calibrated radiation method under different management and meteorological conditions. However, considering the methods data requirements, it appears that those that only require values of greenhouse

transmissivity and daily temperature (Hargreaves) or daily solar radiation (the locally-calibrated radiation approach) may be the most practical to determine ET_o in low-cost plastic greenhouses under standard management conditions in Mediterranean climatic regions.

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